

TRUNCATION IN MESSAGE-ORIENTED PHONOLOGY: THE CASE OF KOREAN VOCATIVE TRUNCATION



SHIGETO KAWAHARA (KEIO UNIVERSITY)

SEUNGHUN LEE (INTERNATIONAL CHRISTIAN UNIVERSITY)

The Role of Predictability
in Shaping Human Language Systems

INTRODUCTION

PREAMBLE

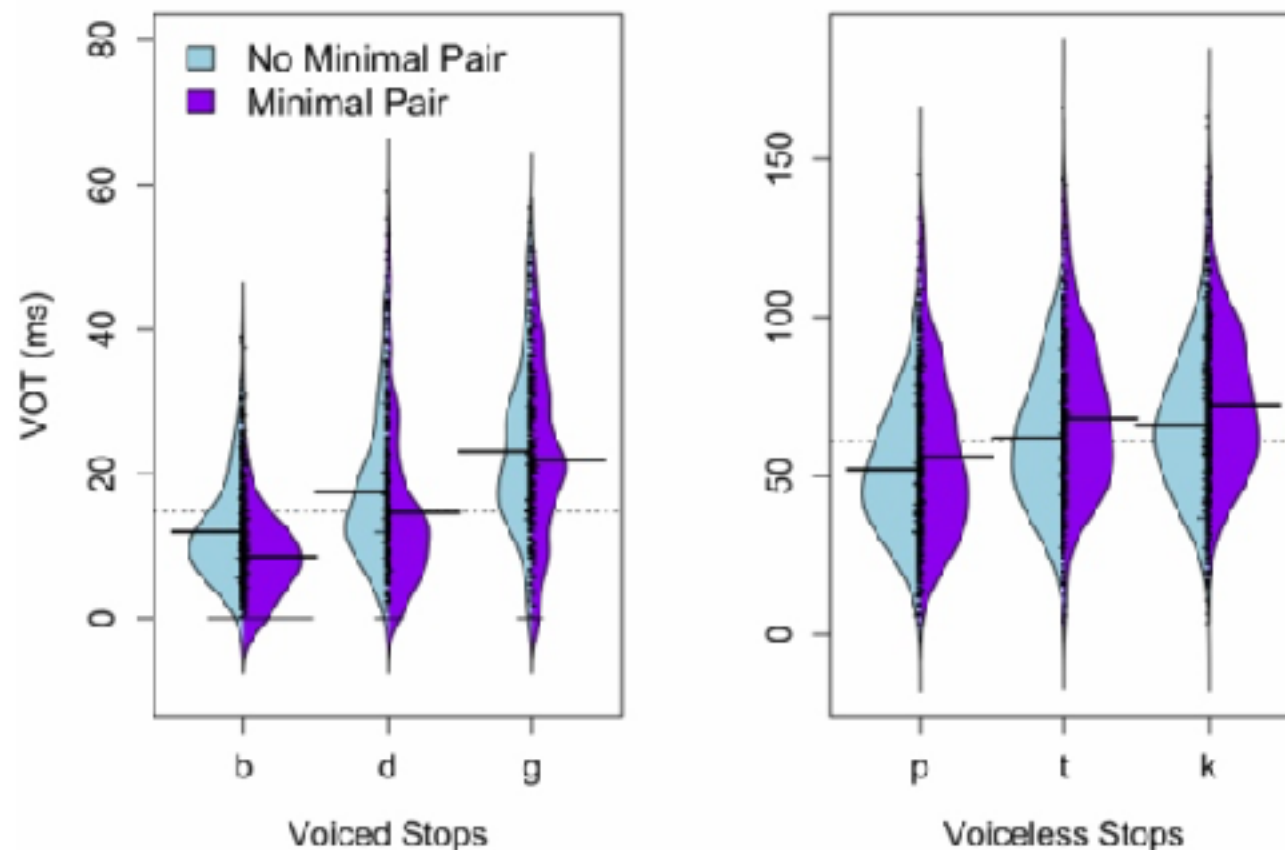
- ❖ Seunghun Lee could not join us this time.
- ❖ I, Shigeto, was a formal phonologist, and interacted with many of them during the graduate time at UMass, and after that at Rutgers. Seunghun has been in a similar situation.
- ❖ I think my role today is to think about MOP from the “viewpoint of formal phonology”.

GOALS

1. Introduce the vocative truncation pattern in Korean which has not been hitherto discussed in the literature.
2. Address the question of “how deep in grammar the MOP principle holds?”
3. Think about why truncation occurs under MOP.
4. Tentatively propose a general “I-Map” theory, which is akin to P-Map and A-Map.

KEY COMPONENTS OF MESSAGE-ORIENTED PHONOLOGY (HALL ET AL 2016)

- ❖ Sound patterns are organized in such a way that it facilitates effective communication.



VOT differences are exaggerated when there is a minimal pair.

From Wedel et al. (2016)

Figure 3: Plots of VOT by minimal pair competitor existence for voiced and voiceless stops. Black horizontal lines within the plots show the mean VOT. The horizontal dotted lines in each plot indicate the group means for all three places of articulation.

HUME, KAWAHARA, SHAW

- ❖ For successful communication, $P(\text{message} | \text{signal}, \text{context})$ needs to be kept high.
- ❖ It is the probability of the listener being able to retrieve the intended **message** given the incoming acoustic **signal** and the **context**.

OUR CONCERN

- ❖ We believe that evidence for MOP for “phonetic-y” patterns is strong, if not undebatable.
- ❖ What an imaginary hard-core generative linguist would say: “what MOP is dealing with is phonetic implementation patterns, and has nothing to do with phonology.”
- ❖ “MOP seems to be a theory of performance, not competence.”
- ❖ “After all, Chomsky (1995 and elsewhere) says that language is not designed for communication.”

OUR GOAL 1

- ❖ We think that phonetics is competence (Keating 1988; Kingston & Diehl 1994), but not every generative linguist would agree.
- ❖ “a condition on phonetic representation is that each symbol be interpreted terms of articulatory and perceptual mechanisms in a **language-invariant manner**: a representation that lacks this property is simply not considered a phonetic representation” (Chomsky 1995, p. 151). (Emphasis added)
- ❖ We can give them a benefit of the doubt and address the following question: “how deep” the principle of effective communication holds?

- ❖ More neutrally put, when “phonologization” occurs, is the linguistic system liberated from the principle of effective communication?
- ❖ Following Shaw et al. (2014), which heavily inspired our work, we show that the principle of effective communication is at work at the morpho-phonological level (see also Kawahara 2016).
- ❖ One bonus: most if not all work on informativity on linguistic patterning is based on English and other Indo-European languages. Our case study is based on Korean, a language that has not been explored from the informativity viewpoint.

OUR GOAL 2

- ❖ Why does truncation occur in the first place?
- ❖ One way to achieve effective communication is to remove **un**informative redundancy from the signal—this is one core claim of MOP, inspired by Shannon (1948).
- ❖ The Korean vocative truncation pattern exactly does this.

CHINESE TRUNCATION: SHAW ET AL (2014)

Truncation examples

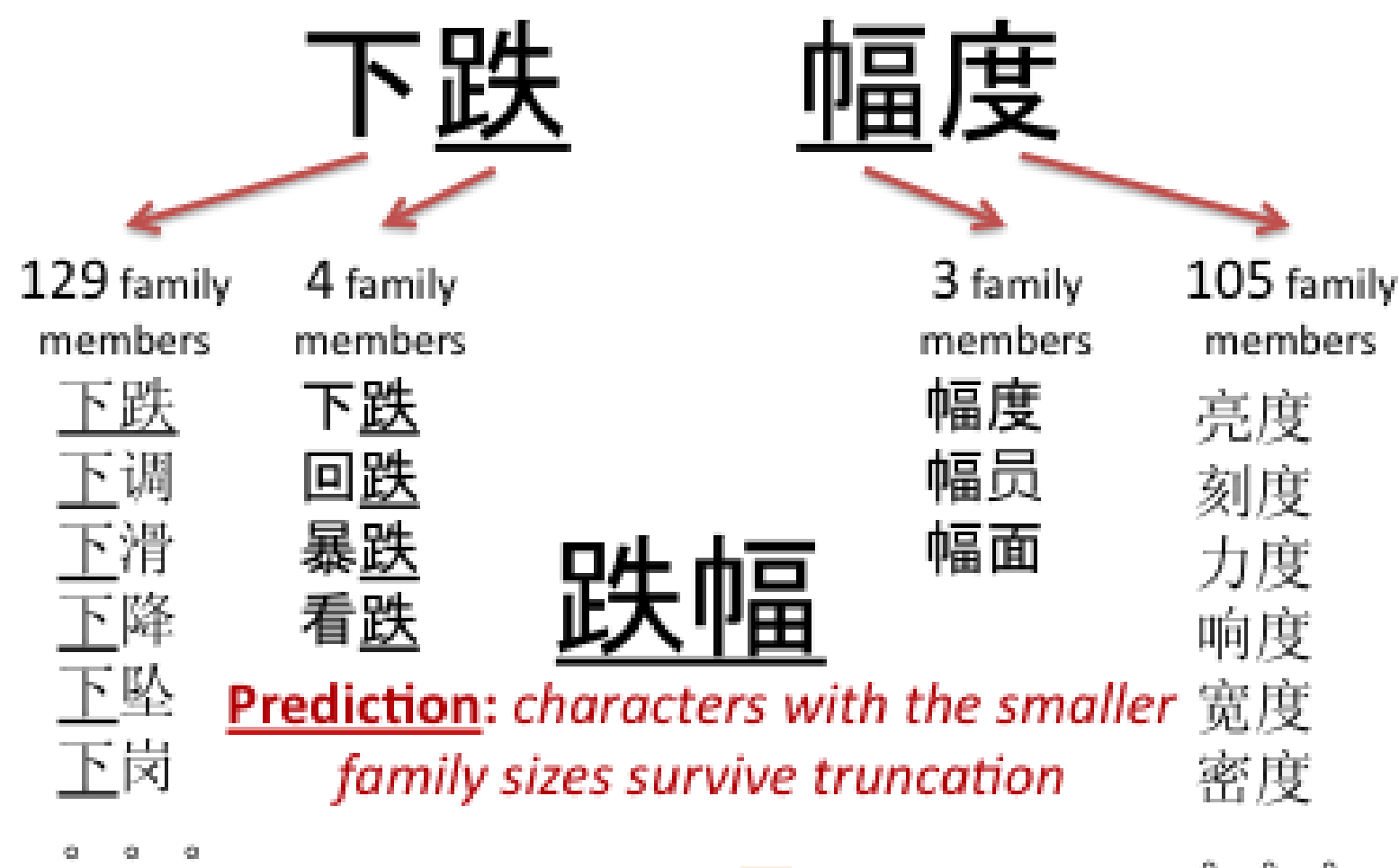
A₁A₂ (46%)	<u>kuòdà</u> 'expand' 'big' 'enlarge'	+	<u>zhāoshōu</u> 'beckon' 'bring in' 'recruit'	=	<u>kuòzhāo</u> 'expand' 'bring in' 'expand admission'
A₁B₂ (10%)	<u>shēnqǐng</u> 'state' 'request' 'application'	+	<u>jǔbàn</u> 'lift' 'do' 'conduct (activities)'	=	<u>shēnbàn</u> 'state' 'do' 'bid for (something)'
B₁A₂ (26%)	<u>xiàdiē</u> 'down' 'fall' 'fall'	+	<u>fúdù</u> 'width' 'degree' 'range'	=	<u>diēfú</u> 'fall' 'width' 'margin of decline'
B₁B₂ (17%)	<u>zhōngyāng</u> 'center' 'core' 'center'	+	<u>yínháng</u> 'silver' 'row' 'bank'	=	<u>yāngháng</u> 'core' 'row' 'central bank'

What determines the syllable that surfaces?

CHINESE TRUNCATION: SHAW ET AL (2014)

Family constituent size

= The number of two character lexical items that share the left or right character



KOREAN VOCATIVE TRUNCATION

STRUCTURE OF KOREAN FIRST NAMES

- ❖ Korean first names consist of two parts: **one part** is shared by the same generation of siblings and cousins, and **the other part** is unique to each person. The order between these two elements alternates between one generation to the other.

Generation I		Generation II		Generation III		
hong+ <u>hui</u>	[hon.hɰi]	<u>jae</u> +eun	[tɕe.ɰn]	min+ <u>su</u>	[min.su]	
dong+ <u>hui</u>	[don.hɰi]	<u>jae</u> +young	[tɕe.jʌŋ]	in+ <u>su</u>	[in.su]	
seok+ <u>hui</u>	[sɔk.hɰi]	<u>jae</u> +hun	[tɕe.hun]	hui+ <u>su</u>	[hi.su]	
yang+ <u>hui</u>	[jaŋ.hɰi]	<u>jae</u> +jun	[tɕe.dʒun]	...		
ja+ <u>hui</u>	[tɕa.hɰi]	<u>jae</u> +u	[tɕe.u]			

VOCATIVE TRUNCATION

What happens in vocative truncation is to take **the unique part** of the names, and add [(j)a] to them. [j] appears when the unique part ends with a vowel. The glide insertion occurs to avoid hiatus.

honga	[hon̩.a]	euna	[ɛ̃.na]	mina	[mi.na]
donga	[don̩.a]	younga	[jʌ̃ŋ.a]	ina	[i.na]
seoka	[sɤ̃.ga]	huna	[hu.na]	huija	[hi.ja]
jaya	[t͡ʃa.ja]	uya	[u.ja]		

TWO REASONS TO THINK THAT THIS PATTERN IS GRAMMATICAL

- ❖ This truncation pattern is a grammatical process, rather than “a random social convention” or “a matter that can somehow be relegated to performance”.
- ❖ Cf. the previously discussed potential criticism against MOP from an imaginary generative phonologist.
- ❖ Vowel sequences are resolved by inserting a glide.
- ❖ The truncation pattern is as phonological as other hiatus resolution patterns (e.g. Casali’s 1996 typological work).

ANOTHER PARALLEL WITH GRAMMATICAL PATTERNS

- ❖ In the vocative truncation pattern, when you meet a new person, and do not know which part of the name is the unique portion, truncation is impossible.
- ❖ This blockage of truncation, sometimes known as “M-Parse Effect”, is very similar to blockage of morphological derivation found in other languages.
- ❖ E.g., Norwegian imperative gaps (Rice 2007: 202-203).

(13) *Well-formed Norwegian imperatives*

- (a) å spise – spis! ‘(to) eat’
- (b) å snakke – snakk! ‘(to) talk’
- (c) å løfte – løft! ‘(to) lift’

Imperative formed via final schwa deletion

(14) *Ill-formed Norwegian imperatives*

- (a) å åpne – *åpn! ‘open’
- (b) å padle – *padl! ‘paddle’
- (c) å sykle – *sykl! ‘bike’

Blocked when it would result in rising sonority sequence

THE CROSS-LINGUISTIC LIST FROM RICE (2007)

- ❖ Turkish suffixation
- ❖ Swedish neuter adjectives
- ❖ Hungarian CCik verbs
- ❖ Reduplication in Mandarin Chinese
- ❖ Hebrew plural
- ❖ Tagalog infixation
- ❖ English schm-reduplication

ANALYSIS OF KOREAN TRUNCATION

Intuition: Vocative forms are used by the family members. The **unique part** of the name, rather than **the generation marker**, is more informative in distinguishing who is being referred to.

Formally: $P(\text{message} \mid \text{signal}, \text{context})$ is the probability of the listener retrieving the correct message given its signal and context (Hall et al. 2016; Shannon 1948).

Let the **context** be the **hui** generation group. Let us suppose that **the intended message** is /hong-hui/.

❖ $P(\text{hong-hui} \mid \{\text{hui}\}, \text{hui}) = 1/5 = 0.2$

❖ $P(\text{hong-hui} \mid \{\text{hong}\}, \text{hui}) = 1$

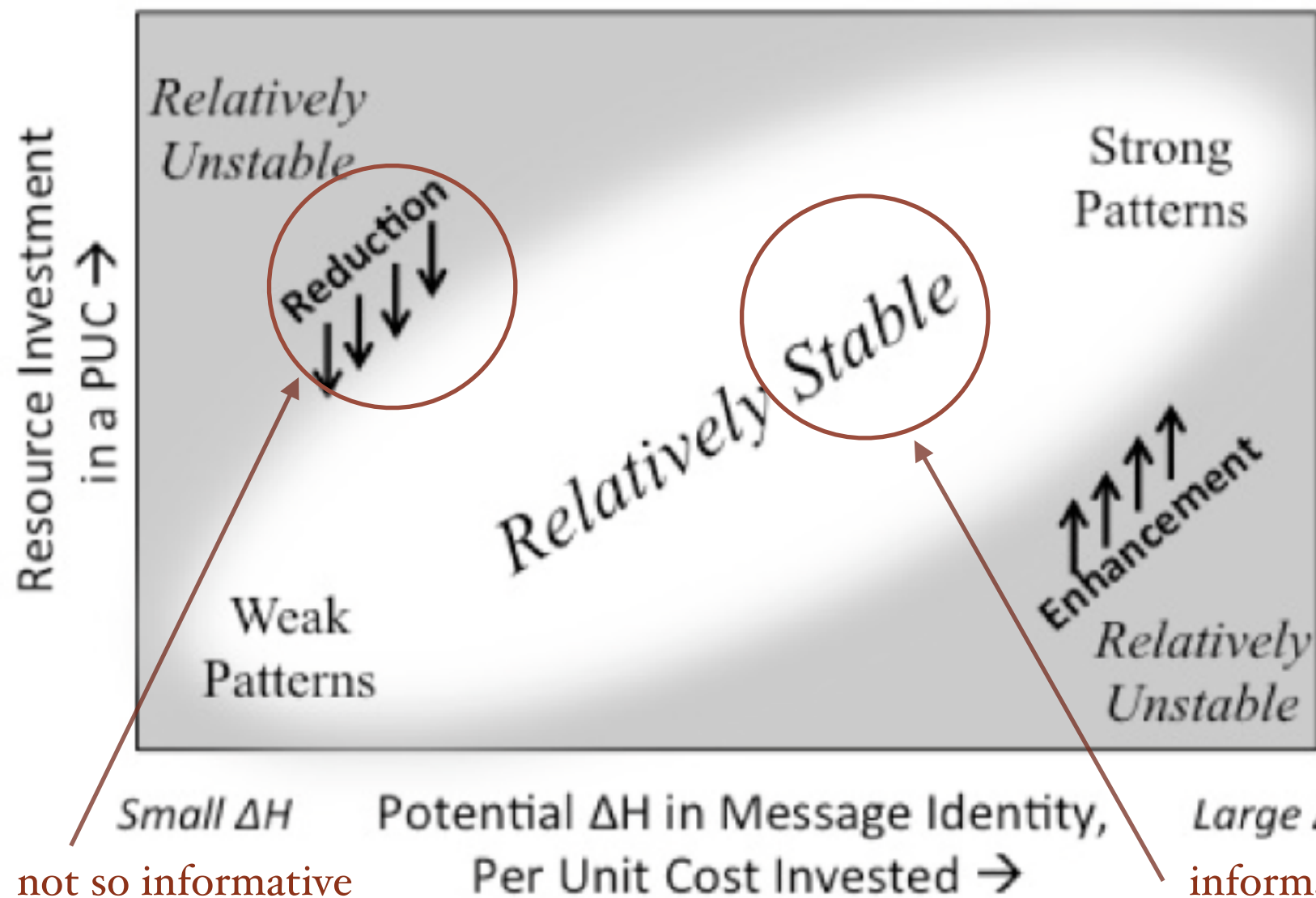


IN TERMS OF ENTROPY

- ❖ We can also cast these differences in terms of entropy. Given the **hui** family,
 - ❖ $P(\text{hui}|\text{hui})=1$, its entropy = $-\log_2(1) = 0$ bit
 - ❖ $P(\text{hong}|\text{hui})=0.2$, its entropy = $-\log_2(0.2) = 2.3$ bits
- ❖ /hong/ has higher $P(\text{message}|\text{signal, context})$ and higher entropy than /hui/.
- ❖ A similar, but related observation, “why do we use our first name rather than last names within a family?” Cf., in Icelandic, people use first names everywhere, because their last names are based on somebody else (their fathers’s first name+son).

- ❖ Our analysis of Korean shows that the context of the message is crucial in determining the informativity.
- ❖ As far as we know, no generative theories of phonology take this sort of social context into account (“family-specific phonology”?).
- ❖ But can perception-driven vowel harmony (Walker 2011) patterns be changed under noisy circumstances? (p.c. Jason Shaw); John McCarthy has asked me a similar question on the phonetically-driven phonetic approach in general.

CAST IN TERMS OF MOP



not so informative

generation marker

informative

the unique part

Figure 5: Phonological patterns as emerging from the effective trade-off between accurate message transmission and resource cost

MORE ON MOP

- ❖ The Korean vocative truncation pattern is exactly what is predicted under MOP.
- ❖ Portions that are not informative are deleted possibly because it is not worth the cost, no matter how the cost is calculated (the time it takes to pronounce it; the articulatory energy that would be spent on it, *Struc, etc).

BUT IT ALSO RAISES A QUESTION

- ❖ But it also raises a non-trivial question.
- ❖ Why is it that languages do not delete every segment after “uniqueness point”, where the target word is distinguished from others?
- ❖ So some redundancy is good; how do we quantify this “some good redundancy”?

MORE ON KOREAN TRUNCATION

- ❖ Korean university name truncations; [~~ihwa~~jɔ̃-de], [yɔ̃nse-de], [korjɔ̃-de].
- ❖ However, [sɔ̃u+de] is impossible, because [sɔ̃+de] can either mean [sɔ̃u+de] or [sɔ̃won+de].
- ❖ Korean speakers seem to allow truncation iff $P(\text{message} | \text{signal}, \text{context})$ is (nearly) 1.

COMPARISON WITH OTHER TRUNCATION PATTERNS

- ❖ More “usual” truncation patterns take portions from initial materials:
 - ❖ Japanese: [keiou+daigaku] => [kei-dai]
 - ❖ Korean: [han̩kuk] + [i̯nhen̩] -> [han̩i̯n]
 - ❖ English: SciFi, CogSci, Prof(essor), etc
- ❖ This tendency follows from the fact (?) that morpheme initial materials are generally informative, or important in distinguishing between words (Beckman 1998).
- ❖ In other words, it is the structure of Korean first names that result in an “unusual” truncation pattern.
- ❖ See also Shaw et al. (2014).

COMPARISON WITH OT

- ❖ In OT, truncation occurs in order to satisfy a prosodic templatic requirement (McCarthy & Prince 1993, 1994, 1995).
- ❖ It predicts that results of truncation are all “prosodically good” in some sense. It is the emergence of the unmarked.
- ❖ In MOP, truncation occurs when some portions are not worth producing.
- ❖ OT remains silent about “in what circumstances truncation occurs”. For example, why is it that truncation is so common in nickname formation?
- ❖ MOP answer: because of the shared information. But we are not sure if MOP explains why the outcome of truncation is usually prosodically defined.

FAITHFULNESS AND INFORMATIVELY

- ❖ To recap our basic idea, those elements that have high entropy are unlikely to delete.
- ❖ In OT and other related framework, this proposal can be recast as:
- ❖ $MAX(\text{high entropy}) \gg MAX(\text{low entropy})$

A GENERAL THEORY OF FAITHFULNESS?

- ❖ Kawahara (2016) proposes the following mechanism:
- ❖ $\text{FAITH}(\text{contrast A}) \gg \text{FAITH}(\text{contrast B})$ if $\text{Entropy}(\text{contrast A}) > \text{Entropy}(\text{contrast B})$.
- ❖ The case at hand is the ranking of faithfulness constraints $\text{IDENT}(\text{voi})_{\text{sing}} \gg \text{IDENT}(\text{voi})_{\text{gem}}$, stemming from $\text{Entropy}(\text{voi})_{\text{sing}} > \text{Entropy}(\text{voi})_{\text{gem}}$.

- ❖ IDENT(voi)sing >> IDENT(voi)gem should hold in Japanese, because geminates are more likely to devoice than singletons due to OCP(voice); e.g., /**doggu/** => [**dokku**], but */**bagu/** => [**baku**].
- ❖ In the Japanese lexicon, Entropy(voice)sing=0.80 > Entropy(voice)gem=0.12

- ❖ This proposal also may make the right prediction about DEP (“No epenthesis”):
- ❖ DEP(high information) >> DEP(low information).
- ❖ See Hume and Maillot 2013 for the evidence from English and French.

- ❖ The proposal is obviously inspired by Steriade's P-map theory (2001/2009). But in that theory, the ranking of faithfulness constraints is determined based on the magnitude of perceptual changes.
- ❖ Can previous analyses resorting to P-map be recast in terms of Information Theory (as Kawahara 2016 did to Kawahara 2006, cf. Hall 2009)?

OR...

- ❖ There is now another proposal, A(rticulatory)-map (Nyun, Inkelas & Rose 2016).
- ❖ May be there are different “maps” (P-map, A-map, and I-map), which each affect the ranking of faithfulness constraints.
- ❖ But what if they demand conflicting ranking conditions? The Japanese epenthesis patterns may provide an interesting testing ground.

CONCLUSION

- ❖ For successful communication, it is important that the speakers' intention is conveyed accurately to the listener.
- ❖ Formally, $P(\text{message} | \text{signal}, \text{context})$ is kept high.
- ❖ For effective communication, uninformative portions can be reduced.
- ❖ Korean vocative truncation pattern precisely does this.
- ❖ It provides support to MOP from a morphophonological perspective (a la Shaw et al. 2014).
- ❖ It may be possible to combine a more formalistic framework like OT and Information Theory.

ACKNOWLEDGEMENTS

Thanks to Beth Hume and Jason Shaw for their comments and general supports on this project. Shaw et al. (2014) was a major inspiration of this work. Thanks also to JSPS grants #26770147, #26770147, #15F15715, which made this project possible. Comments and questions from the audience at the Japanese/Korean Linguistics conference were also very helpful.